

Project

Report:

Machine

Learning

CW

2025

Submitted

by

-

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1.

Campus

Pulse

1.1

Problem

Statement

The

objective

of

the

CampusPulse

project

is

to

analyze

anonymized

student

survey

data

and

uncover

hidden

factors

influencing

student

life

on

campus,

including

predicting

whether

a

student

is

likely

to

be

in

a

romantic

relationship

based

on

lifestyle

and

academic

behaviors.

1.2

Dataset

Description

The

dataset

provided

for

the

CampusPulse

task

consists

of

real,

anonymized

survey

responses

from

students

at

IIT

Guwahati.

The

data

aims

to

capture

a

holistic

view

of

student

life

by

collecting

information

across

three

major

dimensions:

1.

Behavioral

Patterns

Includes

data

on

daily
routines,
such
as
sleep
duration,
screen
time,
and
physical
activity .
Captures
social
engagement
levels,
frequency
of
club
participation,
and
time
spent
on
social
media.

2.

Academic

Attributes

Includes

GPA,

hours

spent

studying

per

week,

class

attendance,

and

academic

satisfaction.

These

features

help

gauge

a

student's

academic

focus

and

performance

trends.

3.

Lifestyle

and

Wellbeing

Factors

Captures

subjective

responses

such

as

stress

levels,

mental

health

status,

and

work-life

balance.

Some

features

include

quantitative

self-ratings,

while

others

reflect

categorical
choices.

Anonymized

Features

To

protect

student

privacy ,

three

survey

responses

have

been

renamed

as:

Feature_1,

Feature_2,

Feature_3.

These

anonymized

variables

are

integral

to

the

analysis,

and

the

task

requires

identifying

their

likely

meanings

using

exploratory

data

analysis

(EDA)

and

statistical

correlations

.

Target

Variable

In_a_relationship:

A

binary

categorical

variable

with

values:

"Yes"

—

The

student

reported

being

in

a

romantic

relationship

"No"

—

The

student

reported

not

being

in

one

This

variable

serves

as

the

label

for
the
classification.

Data

Quality

Notes:

The

dataset

includes

missing

values

in

several

columns,

likely

due

to

skipped

or

incomplete

responses

in

the

original

survey .

A
data
cleaning
and
imputation
strategy
is
necessary
before
building
any
predictive
models.

1.3

Level

-

wise

Approach

LEVEL

1

Here

the

objective

was

to

identify

the

anonymized

features

by

inferring

their

nature

and

correlation

with

other

features

using

EDA

tools.

Approach-

I

used

correlation

heatmap

first

to

understand

how

the

anonymized

features

relate

to

known

variables

like

grades,

absences,

alcohol,

etc.

Pearson

correlation

was

used

to

measure

linear

relationships.

From

the

heatmap,

positive

correlations

of

the

anonymized

features:

Featur e_1

Corr elations
:

failures:

+0.31,

absences:

+0.12

Featur e_2

Corr elations
:

G1:

+0.26,

G2:

+0.25,

G3:

+0.25

Featur e_3

Corr elations
:

Dalc:

+0.62,

goout:

+0.40

From

the

correlation

heatmap

,

I

identified

which

known

variables

are

most

strongly

correlated

with

each

anonymized

feature.

To

understand

these

relationships

visually

and

support

our

feature

identification,

we

now

plot

targeted

scatterplots

—

one

feature

at

a

time

—

against

the

most

relevant

variables.

Scatter

plots

allow

us

to

visually
explore
and
confirm
the
nature
of
the
relationships
highlighted
by
the
correlation
heatmap.
By
plotting
each
anonymized
feature
against
its
most
strongly
correlated
known
variable,

we
can
observe
the
pattern,
strength,
and
any
potential
outliers
in
the
data,
providing
deeper
insight
beyond
correlation
coefficients.

Histograms
reveal
the
distribution
of
each

feature—showing

its

shape,

spread,

and

presence

of

outliers.

Understanding

whether

a

feature

is

symmetric,

skewed,

or

has

multiple

modes

helps

in

selecting

appropriate

imputation

methods

and

informs

further

analysis

decisions.

RESULTS

—

Feature_1

is

most

likely

a

disciplinary

or

academic

risk

score

Feature_2

is

most

likely

an

academic

score

related

to

G1

Feature_3

is

most

likely

related

to

outgoing

nature

or

lifestyle

of

student.

LEVEL

2

The

objective

was

to

find

and

handle

missing

values

of

the

data

by

applying

appropriate

imputation

strategies.

Number

of

missing

values

were

detected

using

`df.isnull().sum()`

These

values

were

imputed

using

the

following

functions:

`df['feature'].fillna(df['feature'].mode()[0])`

```
df['Feature'].fillna(df['Feature'].median())
```

```
df['Feature'].fillna(df['Feature'].mean())
```

For

categorical

and

nominal

data

mode

was

used,

for

ordinal

discrete

data

median

was

used

and

for

numerical

symmetric

data

mean

was

used

to

impute

the

values.

LEVEL

3

Objective:

Ask

at

least

5

insightful

questions

about

the

student

data.

For

each:

-Build

a

meaningful

plot

(bar

chart,

violin

plot,

scatterplot,

etc.)

-Write

a

brief

interpretation

of

the

insight.

Questions

were

framed

based

on

the

relation

of

features

using

correlation

heatmap.

Q1.

Does

parental

education

affect

student' s

academic

performance?

Q2.

Does

weekday

alcohol

consumption

affect

academic

performance?

Q3.

How

does

free

time

influence

going

out?

Q4.

Does

living

lifestyle

affect

student

performance?

Q5.

Does

travel

time

reduce

student' s

free

time?

Box

plot

and

Violin

plot

are

used

to

interpret

these

questions

-

Violin

Plot
shows
the
distribution
shape
of
grades
(`G3`)
for
each
level
of
weekday
alcohol
consumption
(`Dalc`)
and
`failures`.
It
helps
us
understand
density ,
skewness,
and
whether

grades

cluster

or

spread

out.

-

Box

Plot

overlays

key

statistics:

median,

quartiles,

and

outliers,

providing

a

quick

summary

of

central

tendency

and

variability .

LEVEL

4

Objective:

Apply

classification

techniques

to

model

relationship

likelihood

and

assess

their

performance.

Think

critically

about

what

the

models

reveal,

and

what

they

don't.

Here,

I
considered
three
models
-
logistic
regression,
random
forest
and
naive
bayes
as
the
target
variable
was
dichotomous.

After
evaluating
the
performance

I
found
that
logistic

regression

was

best

suitable

for

this

problem

statement.

The

goal

was

to

predict

and

understand

the

feature

influence

on

romantic

relationship.

Naive

bayes

assumes

feature

independence

which

is

why

it

was

not

suitable.

Logistic

regression

provides

coefficients

which

helps

explain

which

factors

increase

or

decrease

the

likelihood

of

being

in

a

relationship.

It
is
often
used
as
a
benchmark
model
in
binary
classification
due
to
reliable
performance.

A
random
forest
model
often
has
a
risk
of
overfitting

and
requires
pruning.
It
also
has
a
slow
training
speed
and
low
interpretability .

The
accuracy
of
the
model
is
62%,
with
F1-score
of
0.69
and
ROC-AUC

of

0.65

LEVEL

5

Interpreting

the

model' s

predictions

was

a

crucial

step

in

understanding

the

underlying

patterns

driving

relationship

outcomes.

By

leveraging

interpretability

techniques

such

as

feature

importance

analysis

and

SHAP

(SHapley

Additive

exPlanations)

values,

the

model' s

decision-making

process

was

made

transparent.

This

allowed

for

the

identification

of

the

most

influential

features

that

contributed

to

predicting

relationship

status,

providing

valuable

insights

into

the

key

factors

affecting

the

outcome.

For

instance,

features

like

sex,

feature1,

etc

emerged

as

significant

predictors,

influencing

the

model's

confidence

in

classification.

Visualizations

such

as

SHAP

summary

plots

and

dependence

plots
were
used
to
illustrate
how
individual
features
affected
predictions
across
different
samples,
enhancing
interpretability
for
both
technical
and
non-technical
stakeholders.
This
interpretability
not
only
boosted

trust

in

the

model

but

also

facilitated

actionable

insights

that

could

guide

further

analysis

or

decision-making

processes

in

real-world

applications.